

Mohammad Javad Ranjbar

École Polytechnique Fédérale de Lausanne (EPFL)

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Research Interests

Multilingual & Low-Resource NLP | Large Language Models | Multimodal AI | Speech Synthesis & Processing

Education

University of Tehran

Tehran, Iran

Master of Science in Computer Software Engineering, GPA: 4.0/4.0 (19.07/20)

2022–2025

Thesis: Zero-Shot Text-to-Speech for Persian Language

Supervisors: Dr. Azadeh Shakery & Dr. Hesham Faili

Curated the largest open Persian speech corpus (4,000+ hours of multi-speaker audiobooks) via a fully automated ASR-and-alignment pipeline; fine-tuned XTTS and Whisper for Persian, achieving 3.6/5.0 naturalness MOS with zero-shot voice cloning.

Amirkabir University of Technology

Tehran, Iran

Bachelor of Science in Electrical Engineering, Last 2 Years GPA: 3.64/4.0

2016–2021

Thesis: Facial Expression Recognition with Robotic Implementation

Supervisors: Dr. M.B. Menhaj & Dr. H. Taheri

Designed and implemented an intelligent personal-assistant system on a low-cost robot with real-time facial emotion recognition, gesture-based responses, and speech command processing.

Publications

2026: M.J. Ranjbar, M. Amini, P. Bathayan, H. Faili, A. Shakery. *PARSA-Bench: A Comprehensive Persian Audio-Language Model Benchmark*. Under Preparation for to EACL 2026 (arXiv:2603.14456).

2026: M.J. Ranjbar, H. Faili, A. Shakery. *ParsVoice: A Large-Scale Multi-Speaker Persian Speech Corpus for Text-to-Speech Synthesis*. Submitted to EMNLP 2026 (arXiv:2510.10774).

2026: M.J. Ranjbar, S. Karimi, A.H. Sheikholeslami, S. Ranjbar Kalahroodi, H. Faili, A. Shakery. *PersianMedQA: Evaluating LLMs on a Persian-English Bilingual Medical Question Answering Benchmark*. Accepted at LREC 2026.

2026: M.J. Ranjbar, H. Faili, A. Shakery. *PersianPunc: A Large-Scale Dataset and BERT-Based Approach for Persian Punctuation Restoration*. Accepted at SilkRoadNLP 2026 (EACL 2026 Workshop).

2026: S. Jafari, T. Azin, M.H. Khan, M.J. Ranjbar, et al. *APARSIN: A Multi-Variety Sentiment and Translation Benchmark for Iranic Languages*. Accepted at SilkRoadNLP 2026 (EACL 2026 Workshop).

2022: M.J. Ranjbar, M.B. Menhaj, H. Taheri. *Social Robots: An Open-Source Framework for Personal Assistant Robots*. ICRoM 2022.

Invited Talks

Jul 2026: *Audio and Text Understanding for Low-Resource Languages*. Invited talk, NLP Seminar, Center for Information and Language Processing (CIS), LMU Munich, Germany.

Technical Skills

- **Deep Learning Frameworks:** PyTorch, TensorFlow, Transformers (HuggingFace), PEFT
- **Programming & Tools:** Python (Expert), R, C++, MATLAB, Git, Docker, L^AT_EX
- **Specialized Libraries:** librosa, ESPnet, fairseq, SpeechBrain, datasets, evaluate
- **Languages:** Persian (Native) – English (Fluent — TOEFL iBT: 103/120 [R: 29, L: 29, S: 23, W: 22])

Professional Experience

EPFL NLP Lab

Lausanne, Switzerland

Research Intern

Mar 2026–Present

Working under the supervision of Dr. Antoine Bosselut on Apertus 2, Switzerland's fully open multilingual LLM: adding native audio understanding, and quantifying the value of synthetic (TTS) speech for low-resource languages via exchange-rate and training-curriculum studies across five languages.

University of Tehran

Research & Teaching Assistant

Tehran, Iran

Jan 2023–Mar 2026

Research Assistant at NLP Lab & Intelligent Information Systems Lab — led research on Persian speech synthesis and medical LLMs.

Chief Teaching Assistant: ML Methods in NLP and Digital Speech Processing — assignments, discussions, and course material development.

Teaching Assistant: Information Retrieval, Machine Learning, Statistical Inference, Deep Learning, Advanced Algorithms — designed final projects and homework for 200+ students across multiple courses.

Sharif University of Technology

Teaching Assistant

Tehran, Iran

Feb 2024–Sep 2025

Machine Learning & Deep Learning Courses — designed and implemented an LLM-agent assignment for the Deep Learning course; organized an ML contest with 200+ participants across NLP, Computer Vision, and General ML, developing both problem statements and complete solution frameworks.

Sofia

Co-founder & NLP Engineer

Tehran, Iran

Aug 2023–Aug 2024

Omnipresent RAG-based Customer Service Chatbot — architected and deployed a multi-platform chatbot (Telegram, WhatsApp, Web) with a custom RAG pipeline; reached **20,000+ active users** on the University of Tehran website.

Neuromatch Summer School

Teaching Assistant

Remote

Summers 2024 & 2025

Mentored international students in Deep Learning with a focus on NLP; guided daily sessions on architectures and applications.

Amirkabir Robotics School (FIRA Academy)

Instructor & Developer

Tehran, Iran

2020–2022

Taught Image Processing & ML courses; developed the SWIMBot platform for Diginext; created a computer-vision and ML curriculum.

Amirkabir University of Technology

Teaching & Research Assistant

Tehran, Iran

2020–2022

TA for Introduction to Computational Intelligence; RA at the Computational Intelligence and Large-Scale Systems Lab, working on facial emotion recognition.

Key Projects

PARSA-Bench: Evaluation of Persian Audio-Language Models: *Research Project – 2026*

Built a benchmark of 8,000+ Persian audio samples across 16 tasks — ASR, translation, intent, NER, QA, emotion, and culturally grounded tasks such as poetry meter (vazn), poetry style (sabk), and Dastgah music — combining real and TTS-generated audio; 10 tasks are the first of their kind in any language.

Multimodal Question-Answering System: *ML Course Contest Design – 2025*

Designed a synthetic multi-object counting dataset of Iranian animal species using the Google Search API and SAM2, and trained a multimodal QA model for animal counting.

Selected Coursework & Certifications

Graduate Courses (GPA 4.0/4.0):

- Machine Learning (20/20)
- Digital Speech Processing (20/20)
- Statistical Inference (19.2/20)
- Deep Learning (18.75/20)

Professional Certifications:

- Machine Learning (Stanford/Coursera)
- NLP Specialization – 4 Courses (DeepLearning.AI)
- Generative AI with LLMs (DeepLearning.AI)
- Algorithms Specialization (Stanford/Coursera)

Honors & Academic Excellence

2022: Top 0.5% (76th among 13,000) – Computer Engineering National Master's Entrance Exam

2022: Top 1% (35th among 2,000) – Computer Science National Master's Entrance Exam

2016: Top 0.25% (411th among 163,000) – National University Entrance Exam

References

Dr. Azadeh Shakery

Associate Professor
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Dr. Hesham Faili

Professor
University of Tehran
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Dr. M.B. Menhaj

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